

Prospective BBQ Nation Party Assignment

1. Coins on a Table

Two players take turns placing identical Rs 1 coins on a rectangular table. No coin may overlap another or hang off the edge. The first player who cannot place a coin loses. Does either player have a guaranteed winning strategy? If so, who — and what is it?

2. Chomp

You have an $m \times n$ chocolate bar. The bottom-left square is poisoned. A move means choosing any remaining square and eating it along with every square above and to the right of it. Whoever is forced to eat the poisoned square loses. Show that for any board larger than 1×1 , the first player can always win. The tricky part: can you prove they win *without* describing the actual moves?

3. Guess Two-Thirds of the Average

Everyone in class secretly writes down a number between 0 and 100. The winner is whoever is closest to two-thirds of the average of all the submitted numbers. What number should you write — and why? (We'll actually collect these and crown a winner, so think hard.)

4. The St. Petersburg Game

I flip a fair coin repeatedly until it first lands heads. If the first heads appears on the n -th flip, I pay you 2^n rupees. The expected payout of this game is *infinite*. So: how much would you genuinely pay me to play it once? And why isn't your answer 'any amount up to infinity'?

5. The Ultimatum

Player A is handed Rs 1000 and proposes how to split it with Player B. If B accepts, they split it exactly as A suggested. If B rejects, *both* get nothing. Suppose A offers B only Rs 10. As B, do you accept? And as A, what split should you actually propose?

6. Matching Pennies

You and an opponent each secretly set a coin to Heads or Tails, then reveal simultaneously. You win if the coins match; your opponent wins if they differ. Is there a 'best' move? What happens once your opponent figures out your habits?

7. The Traveler's Dilemma

Two travelers own identical antiques, which an airline damages. The airline asks each traveler — separately, no talking — to claim a value between Rs 2 and Rs 100. It will reimburse *both* travelers the lower of the two claims. As a kicker: whoever claimed the lower amount gets an extra Rs 2, and whoever claimed higher is docked Rs 2 (if the claims are equal, both just get the claim). What do you write?

8. The Centipede

There is a growing pot of money. On your turn you may *take* — grab the larger share and end the game — or *pass*, which grows the pot but hands the next move to your opponent. The game lasts 6 rounds and the pot doubles each round. When should you take?

9. Two Gambles, Two Rounds

Round 1 — pick one: (A) Rs 1 crore, guaranteed. (B) An 89% chance of Rs 1 crore, a 10% chance of Rs 5 crore, and a 1% chance of nothing.

Round 2 — pick one: (C) An 11% chance of Rs 1 crore, otherwise nothing. (D) A 10% chance of Rs 5 crore, otherwise nothing.

Commit to a choice in *both* rounds. Which did you pick — and are your two choices consistent with each other?

10. The Urn

An urn holds 90 balls. Exactly 30 are red. The other 60 are some unknown mix of black and yellow — it could be any split.

- **Bet 1:** would you rather bet 'the next ball is red' or 'the next ball is black'?
- **Bet 2:** would you rather bet 'red or yellow' or 'black or yellow'?

Most people pick *red* in Bet 1 and *black-or-yellow* in Bet 2. Can a single, consistent set of beliefs about the odds justify both of those choices?

11. How Much to Bet?

You have a coin biased 60/40 in your favour and a bankroll of Rs 100. You may bet any fraction of your current bankroll on each flip, and you can keep betting for an hour. Every bet has positive expected value. What fraction of your bankroll should you stake each time to end up as rich as possible? And what happens if you go all-in every single time?

12. A Taste of Poker

You are dealt a hand that is equally likely to be the *best* possible or the *worst* possible — a coin flip. You can either *check* (go straight to showdown and reveal) or *bet* (your opponent then chooses to call or fold). Should you *ever* bet when you're holding the worst hand?

If you have opened this assignment and read the questions, tell me the name of your favourite band / artist in the chat at the very start of the next session.